

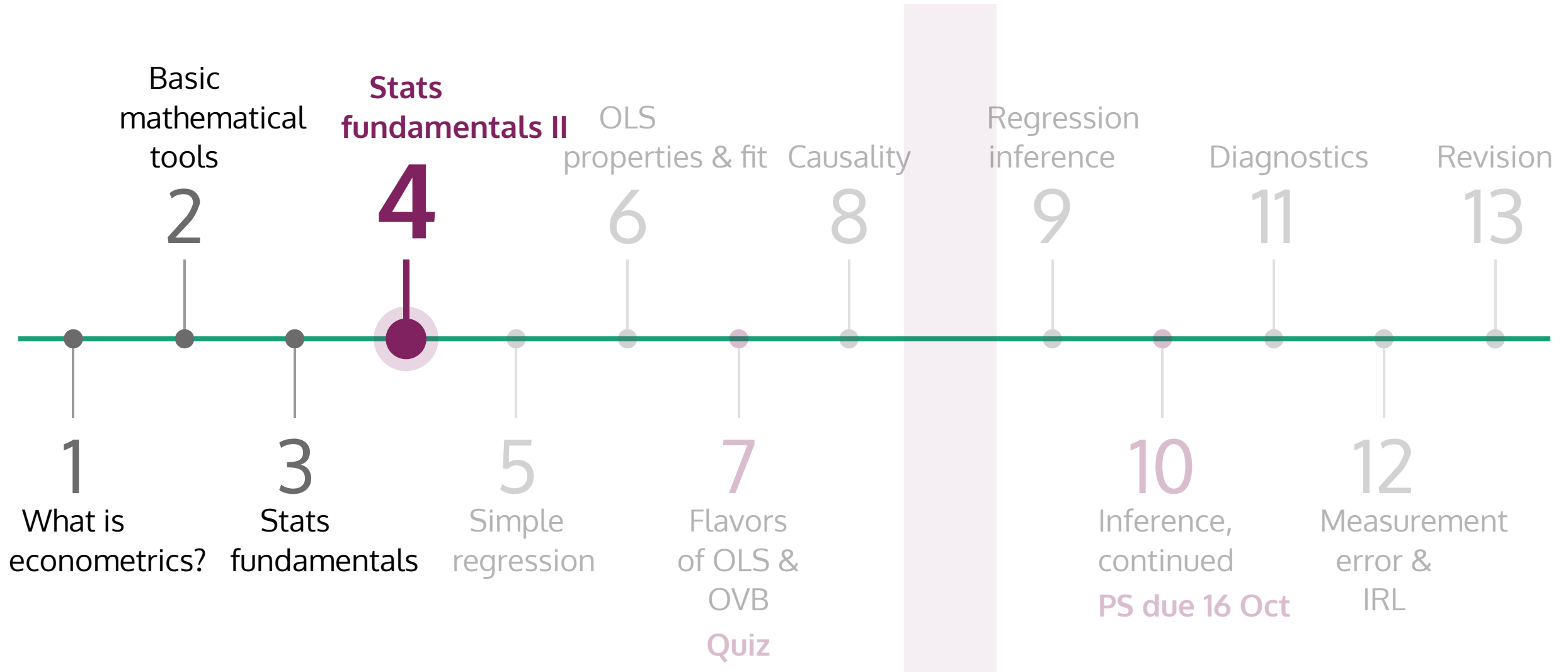


ECON 2041

Introductory Econometrics

Great expectations

Plan for the semester



Plan for today

- Your grade forecasts, revealed
- Income and life expectancy in the US
- Next week: a correlation gallery, and where a covariance comes from

What you expect of yourself

What I asked you

Your own semester

- Expected final mark (0-100)
- Plans:
 - attend live?
 - study hours?
 - try tutorials before class?
- The mark if you did "everything"

Three hypothetical students

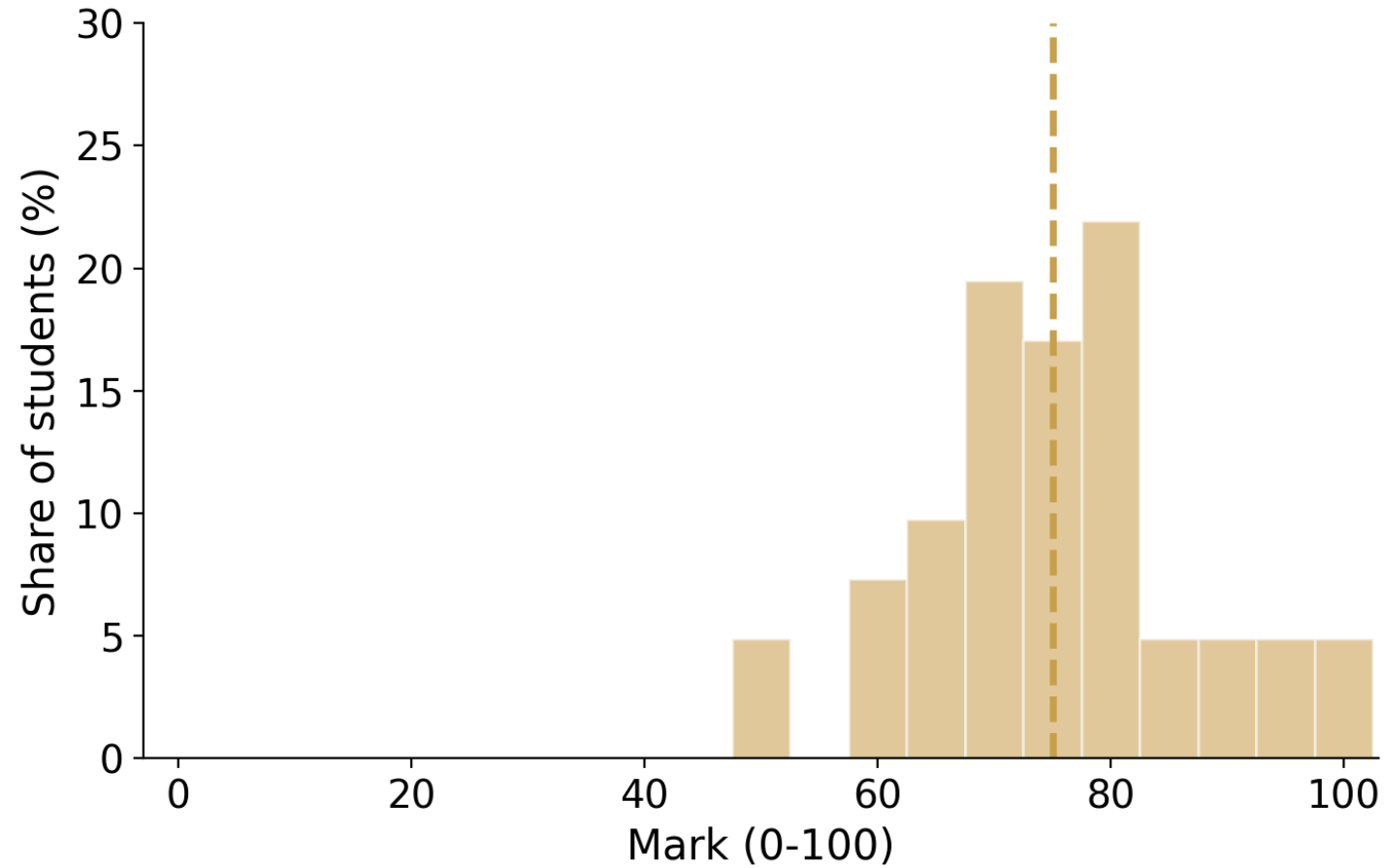
- **Attends live lecture:**
rarely / some weeks / most
- **Study hours:** <2 / 2-5 / 5+
- **Try tutorials first:** no / yes

For each:

- a mark
- % chance they pass,
- % chance they get a D

Nobody expects to fail

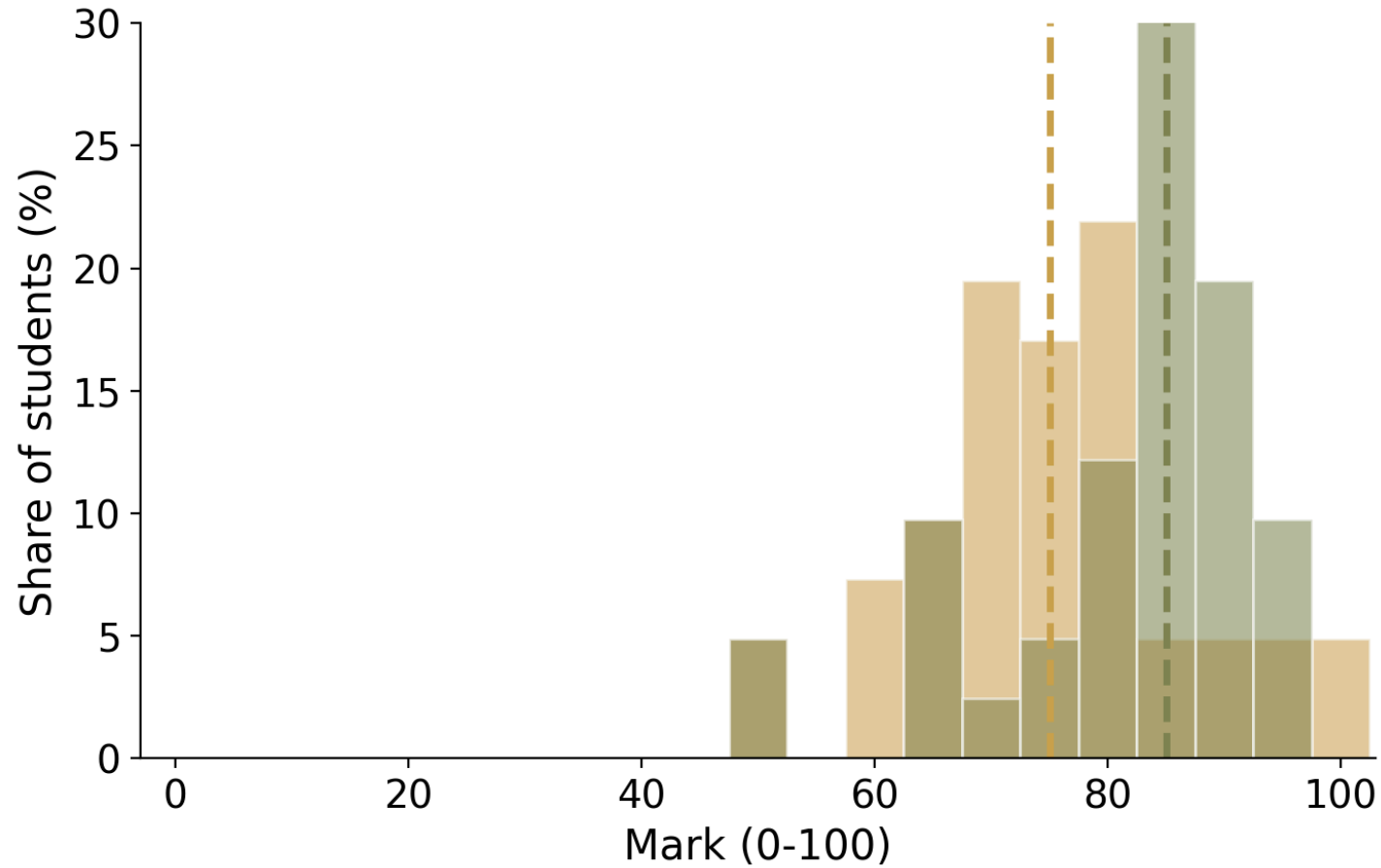
Own forecast, 2026 ($N=41$), median 75



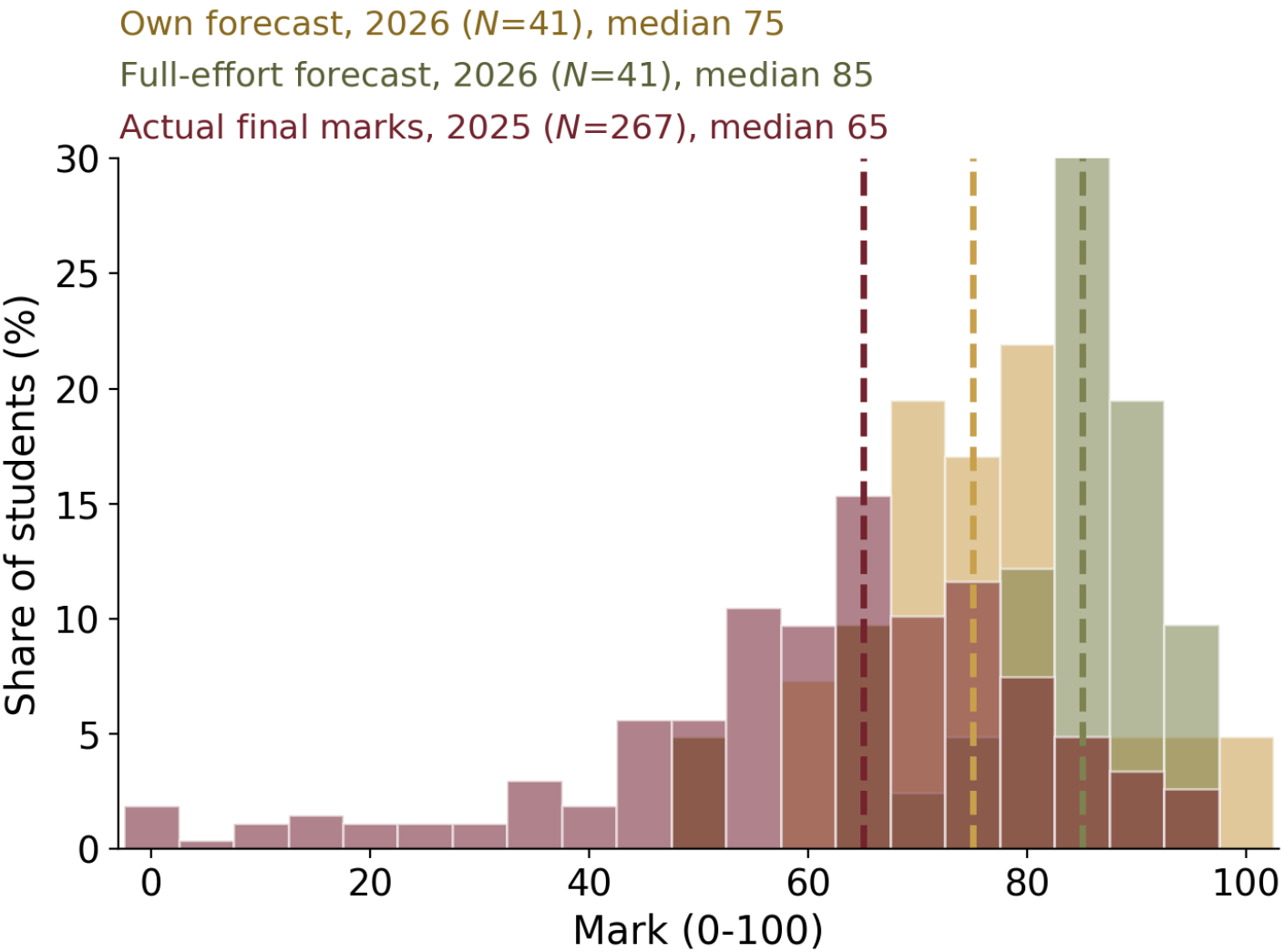
Full effort: you expect it to add about 7 points!

Own forecast, 2026 ($N=41$), median 75

Full-effort forecast, 2026 ($N=41$), median 85

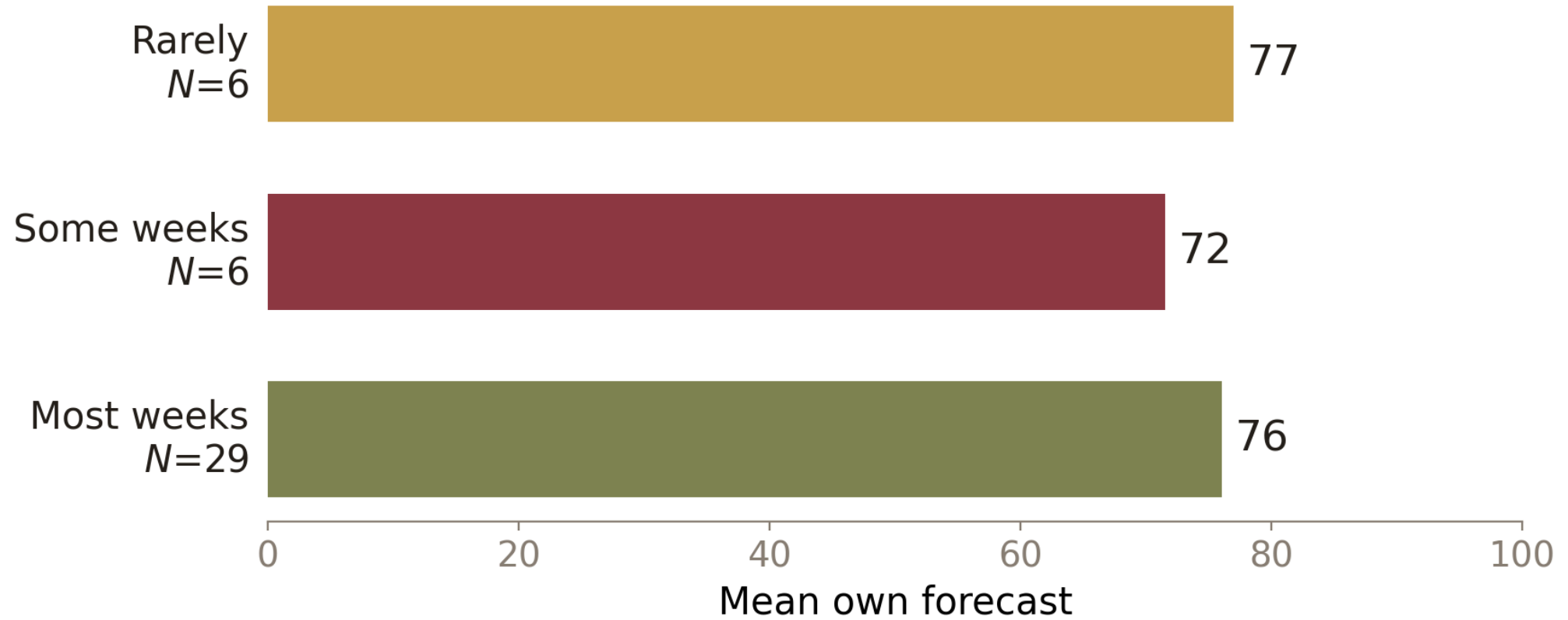


Last year's marks

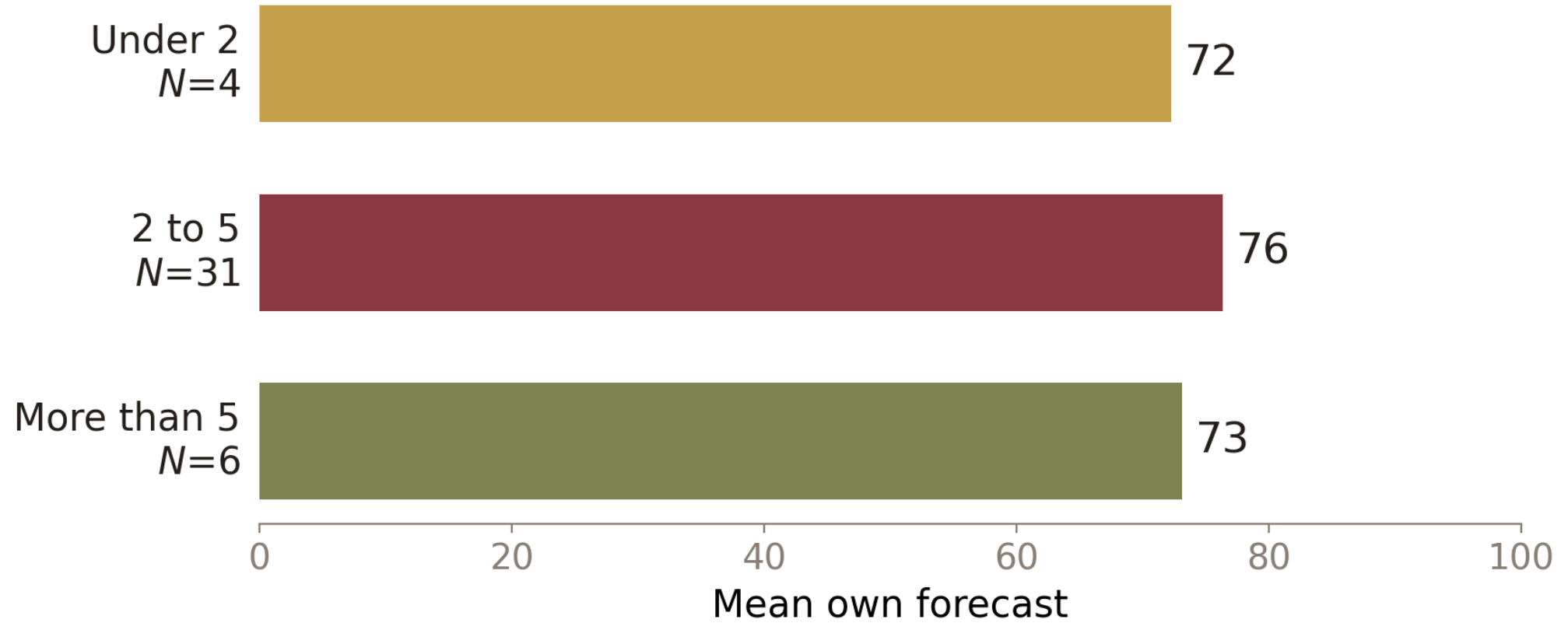


Your forecast, given your plans

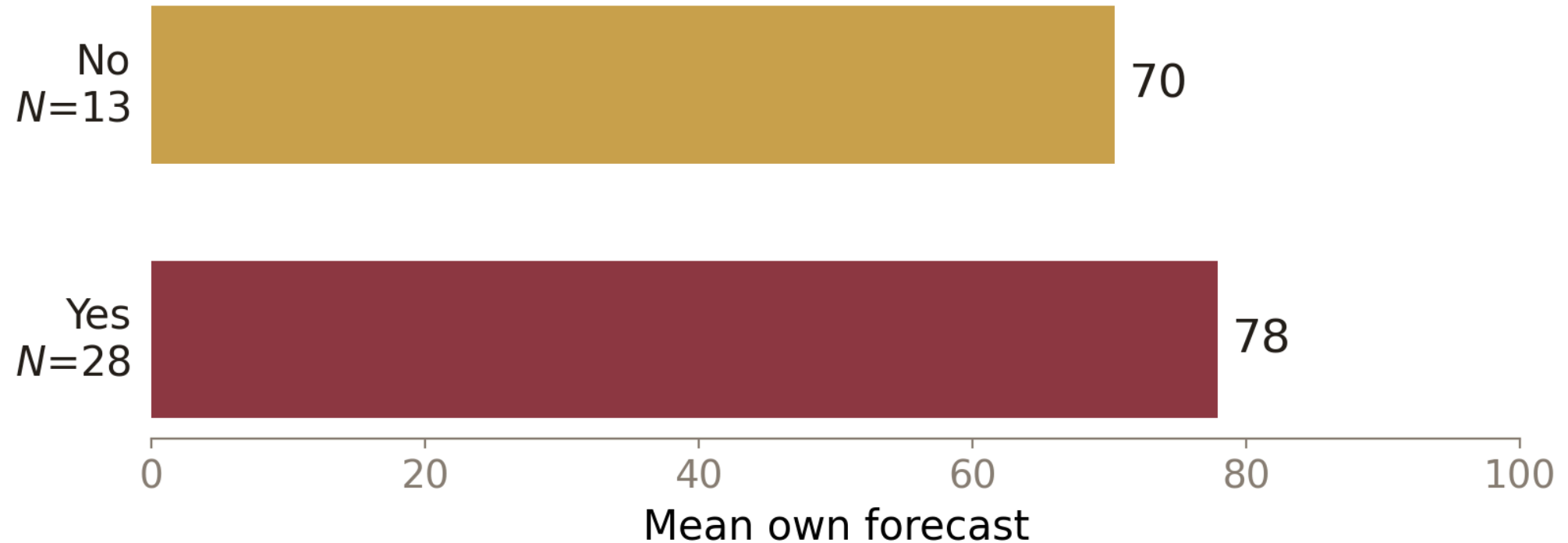
Attendance plans: flat



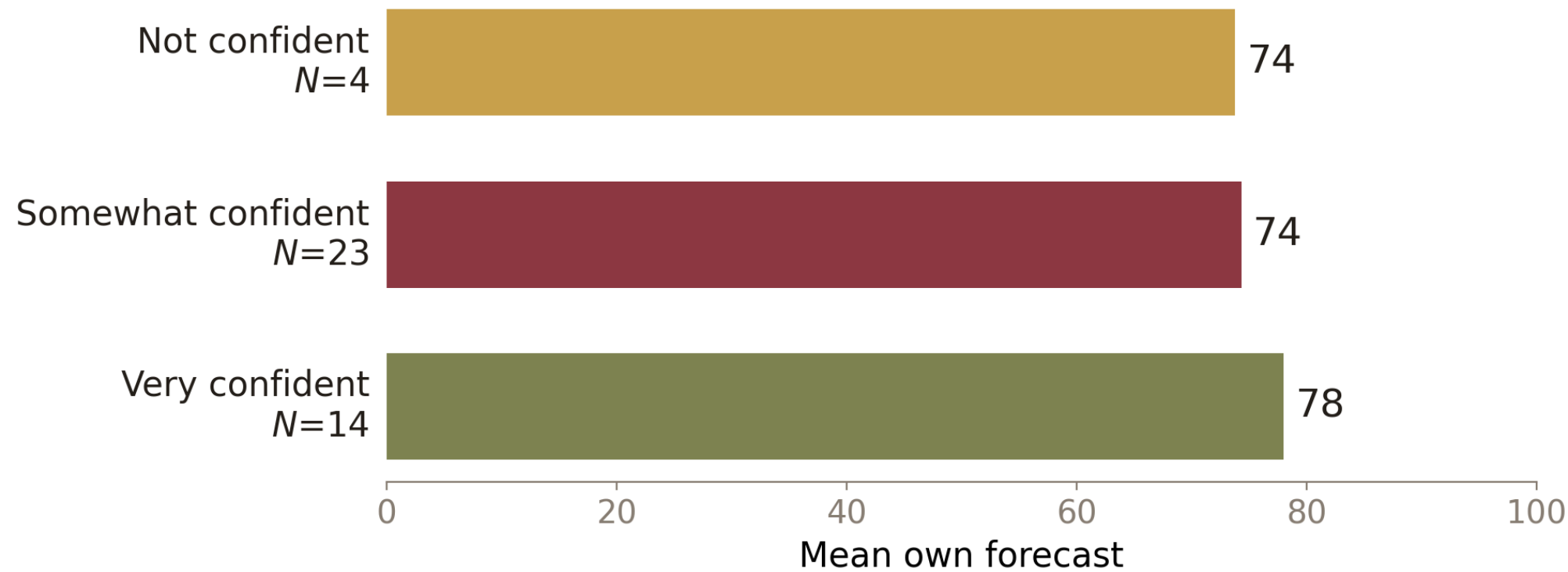
Study-hour plans: flat



Tutorials before class: where the action is

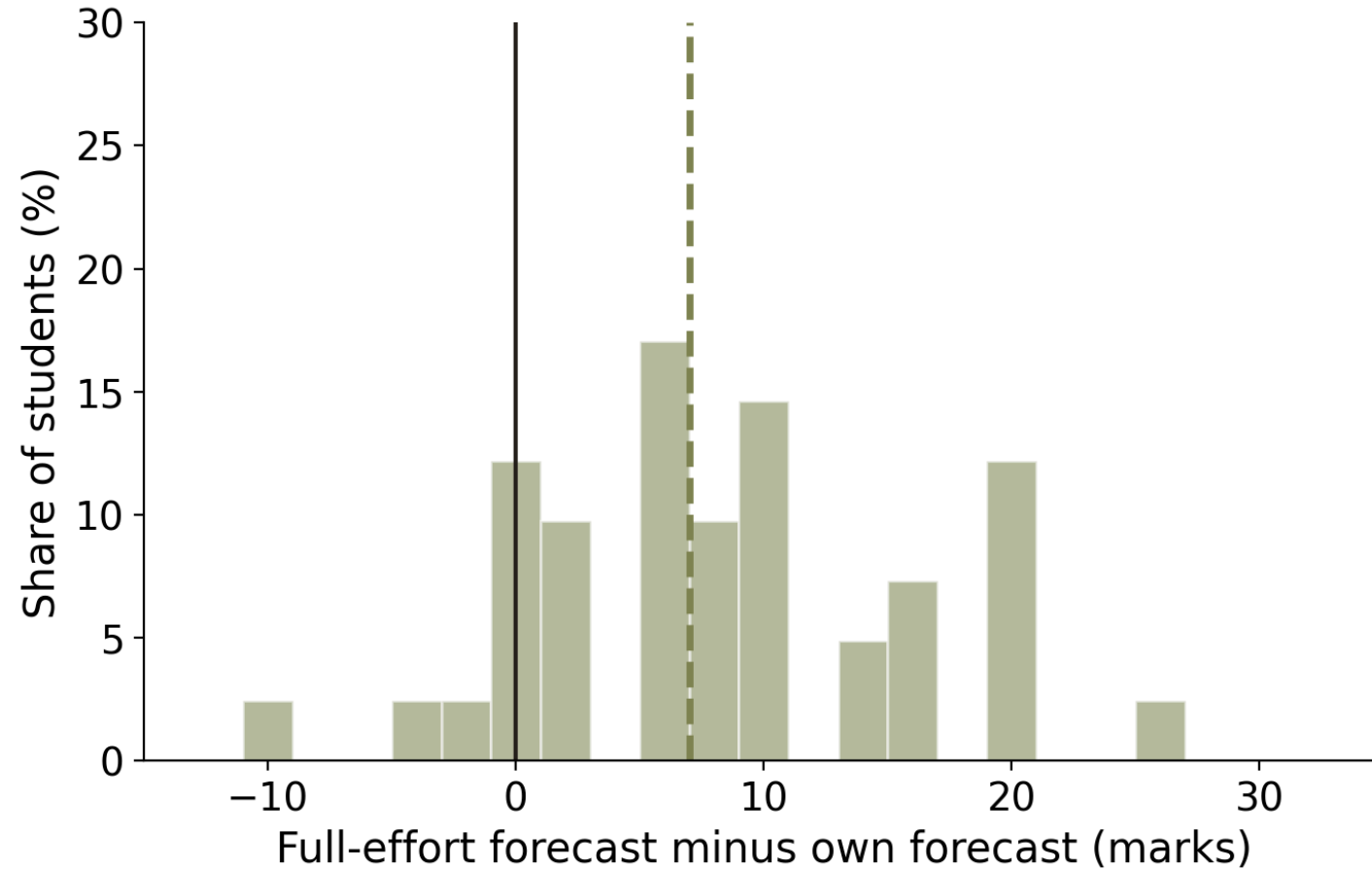


Confidence in the plan barely matters



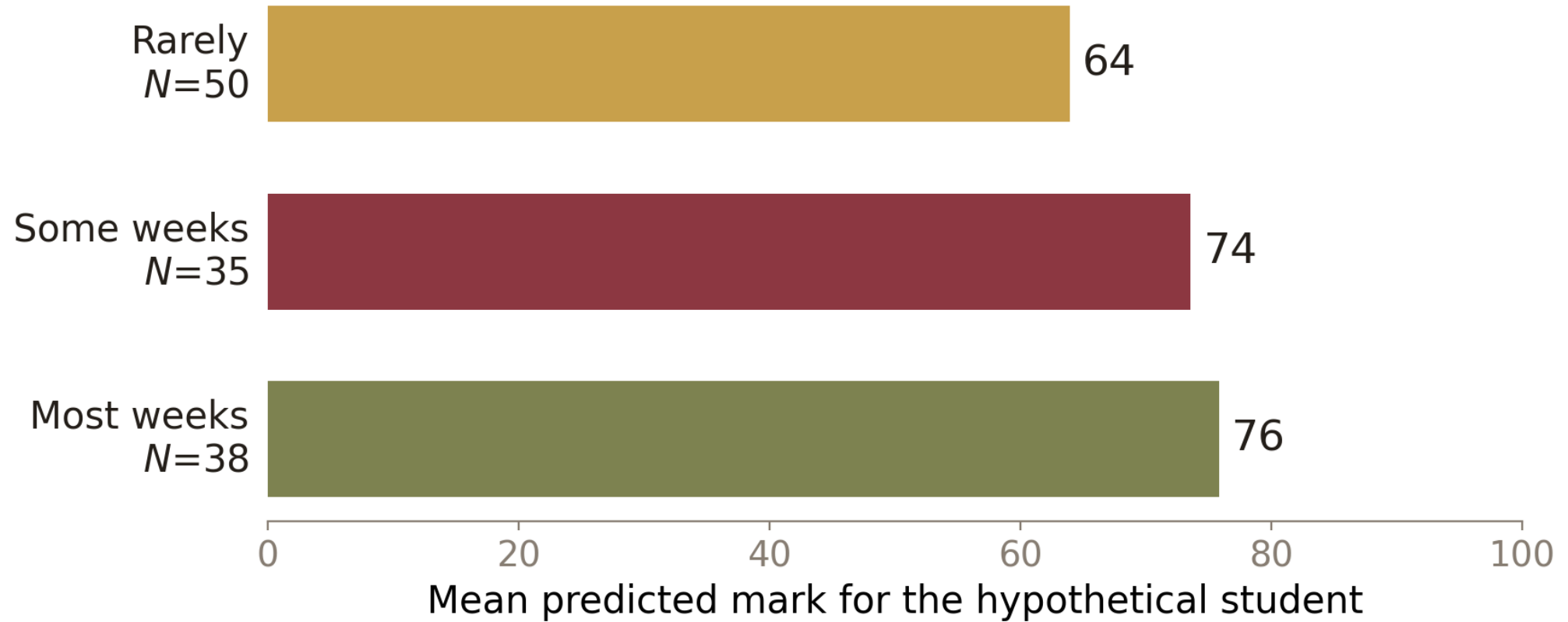
Effort, student by student

Full-effort minus own forecast, 2026 ($N=41$), median 7

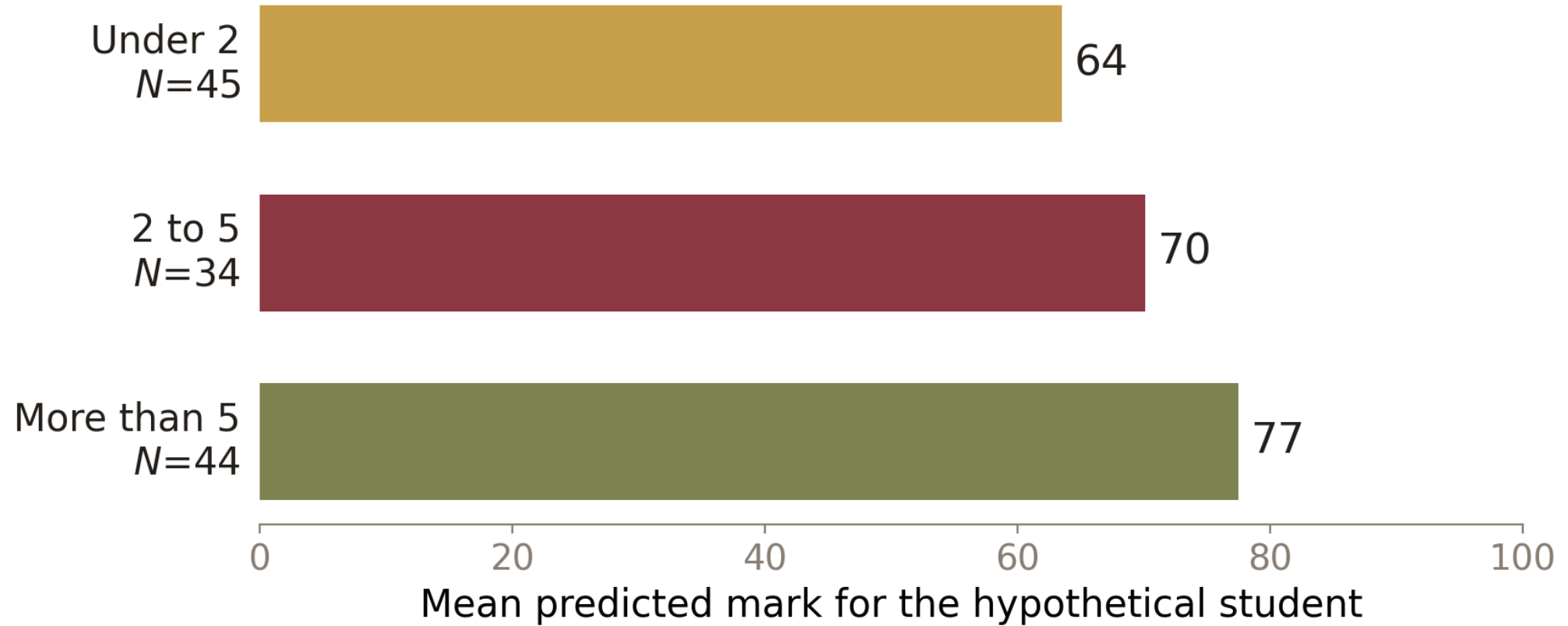


What you expect of others

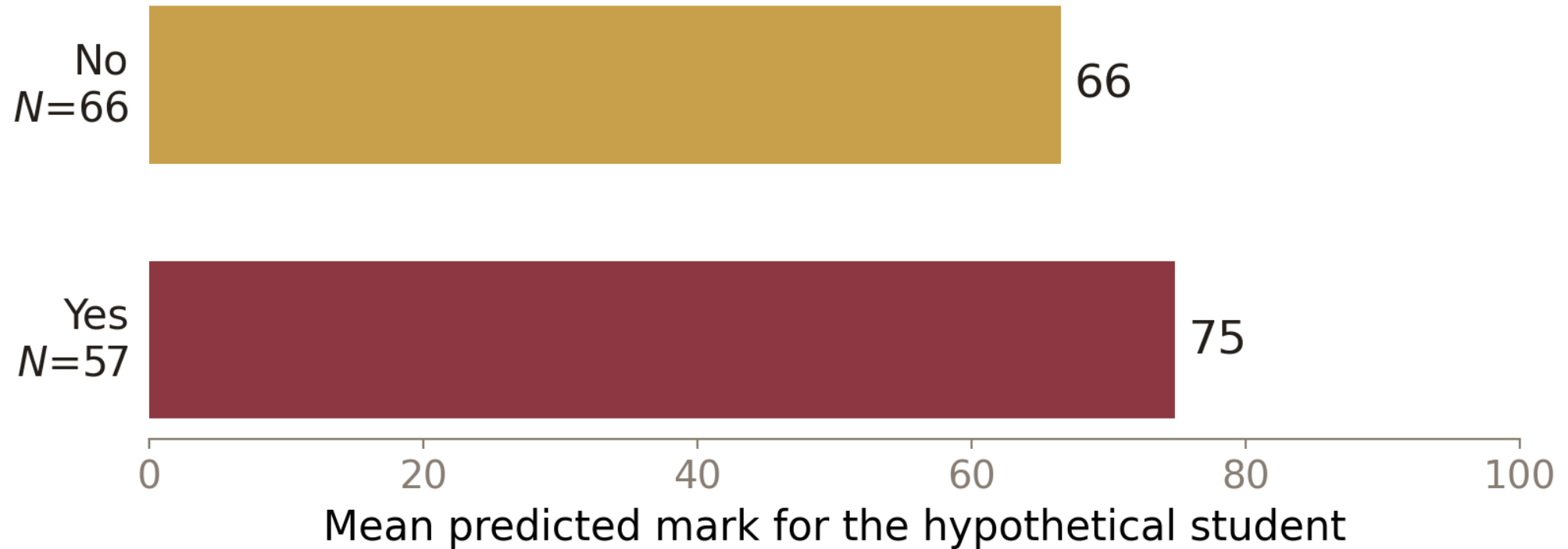
You DO think that attendance matters



You also think that study hours matter



Same story for tutorial work before class



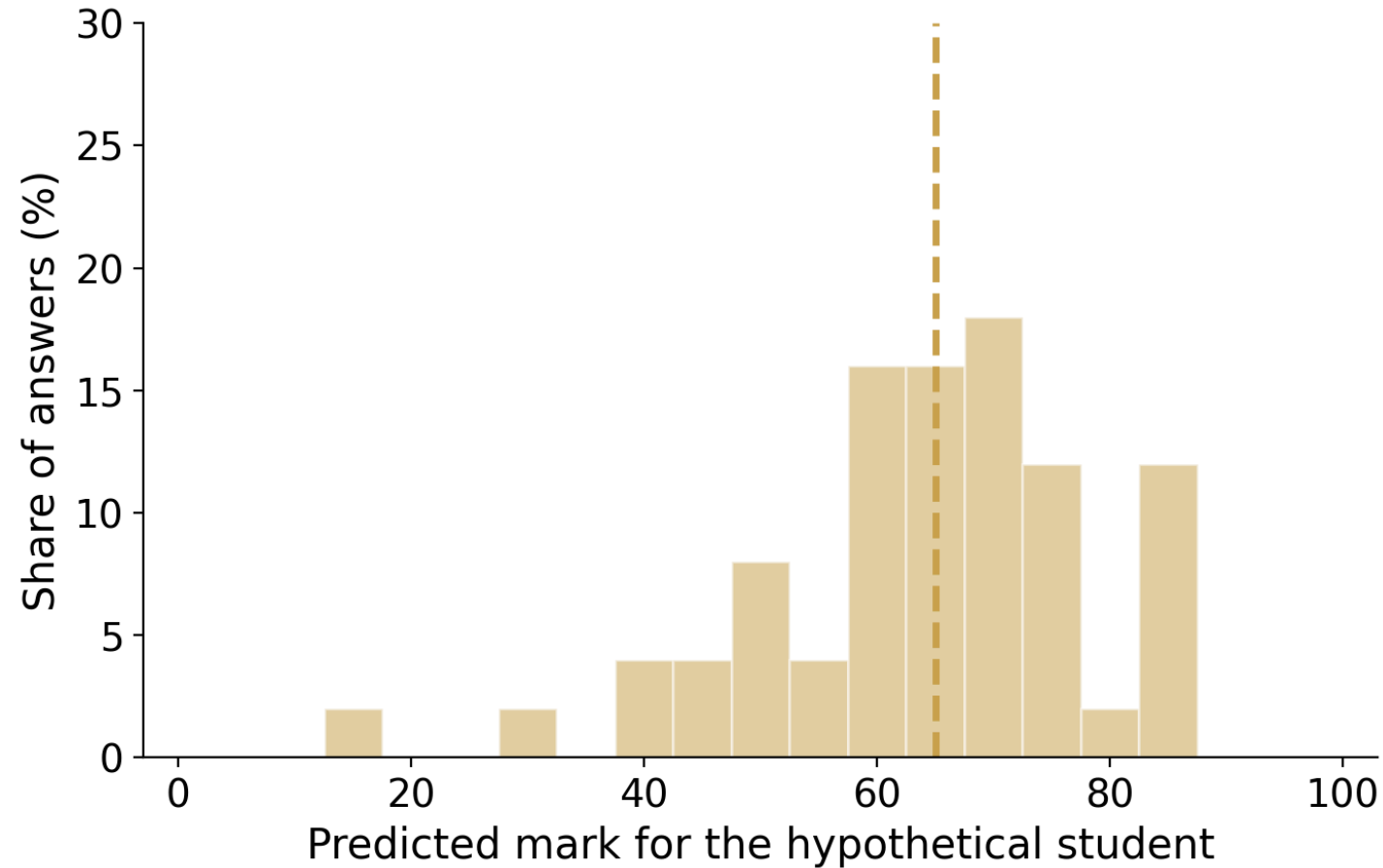
Independence by design

Share of hypothetical students at each study-hours level, within each attendance level

Assigned attendance	Assigned study hours per week			<i>N</i>
	<2	2 - 5	5+	
Rarely	44%	30%	26%	50
Some weeks	20%	26%	54%	35
Most weeks	42%	26%	32%	38
All vignettes	37%	28%	36%	123

Predicted marks, given "rarely"

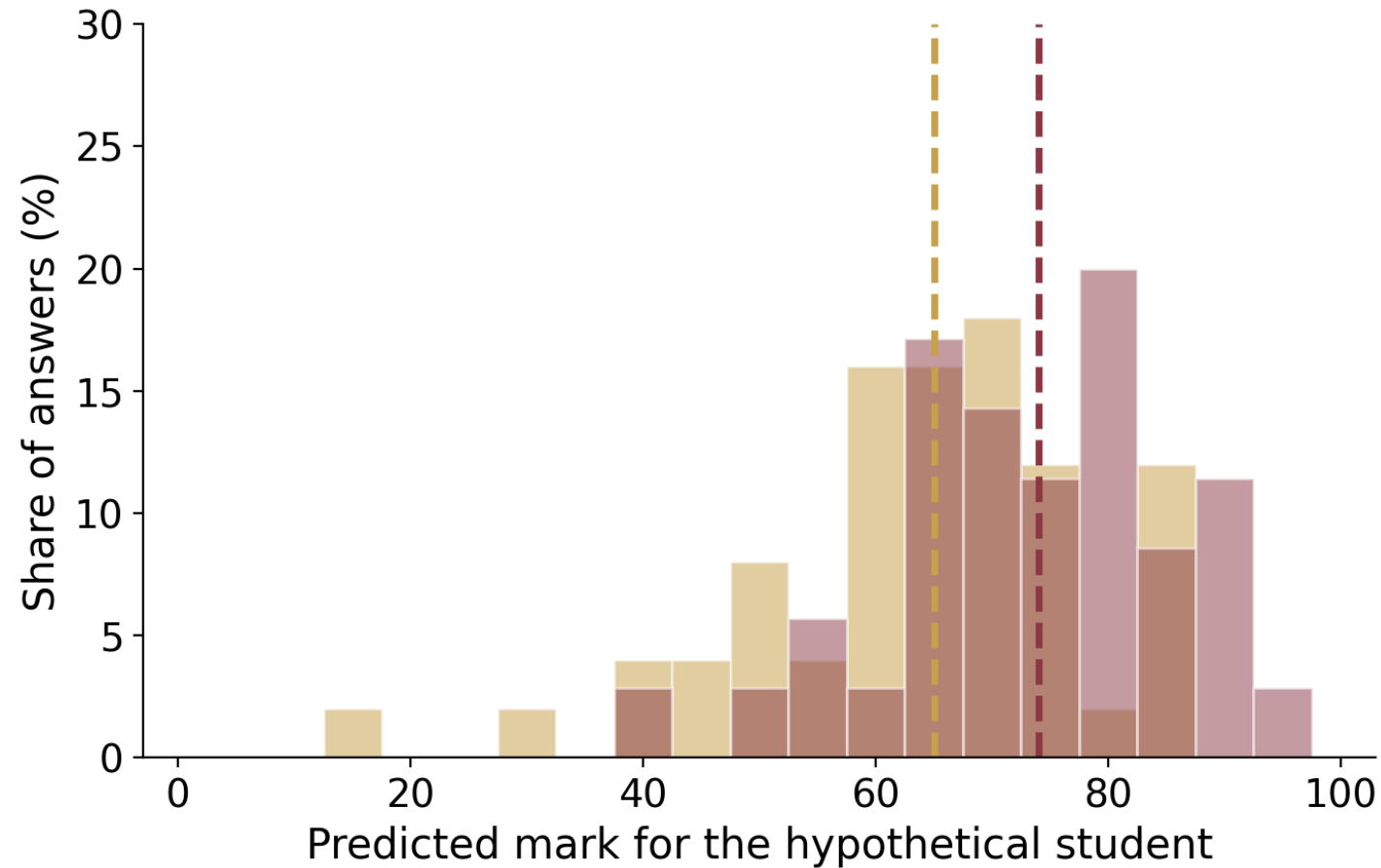
Assigned rarely ($N=50$), median 65



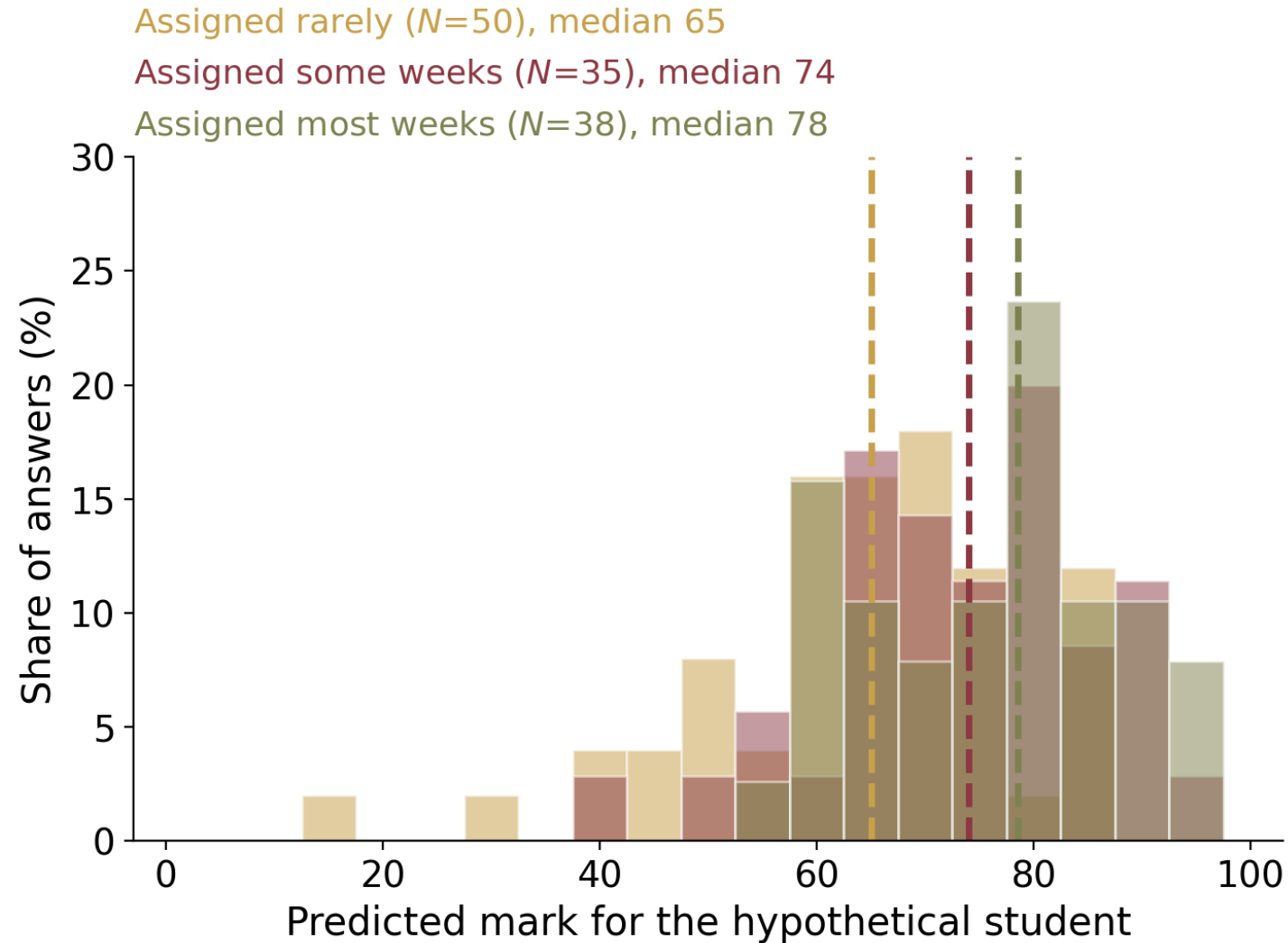
Predicted marks, given "some weeks"

Assigned rarely ($N=50$), median 65

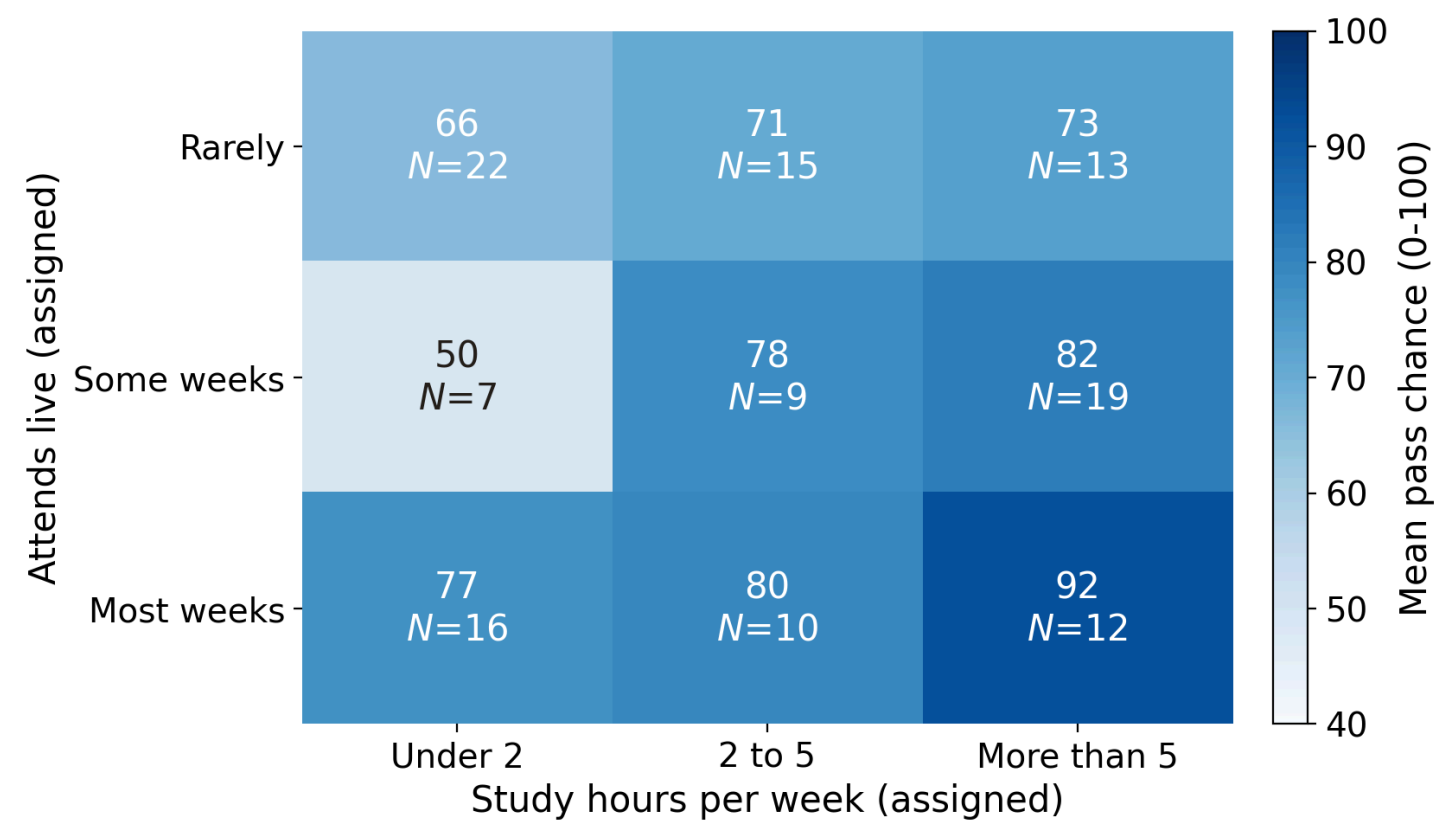
Assigned some weeks ($N=35$), median 74



Marks are not independent of attendance



Pass chance, given attendance & hours



Distinction $\Rightarrow$ pass

For each hypothetical student, two chances:

- $P(\text{mark} \geq 50)$, the chance of a pass
- $P(\text{mark} \geq 75)$, the chance of a distinction

Every mark of 75 or more is also a mark of 50 or more

$\{\text{mark} \geq 75\} \subset \{\text{mark} \geq 50\}$, so $P(\text{mark} \geq 75) \leq P(\text{mark} \geq 50)$

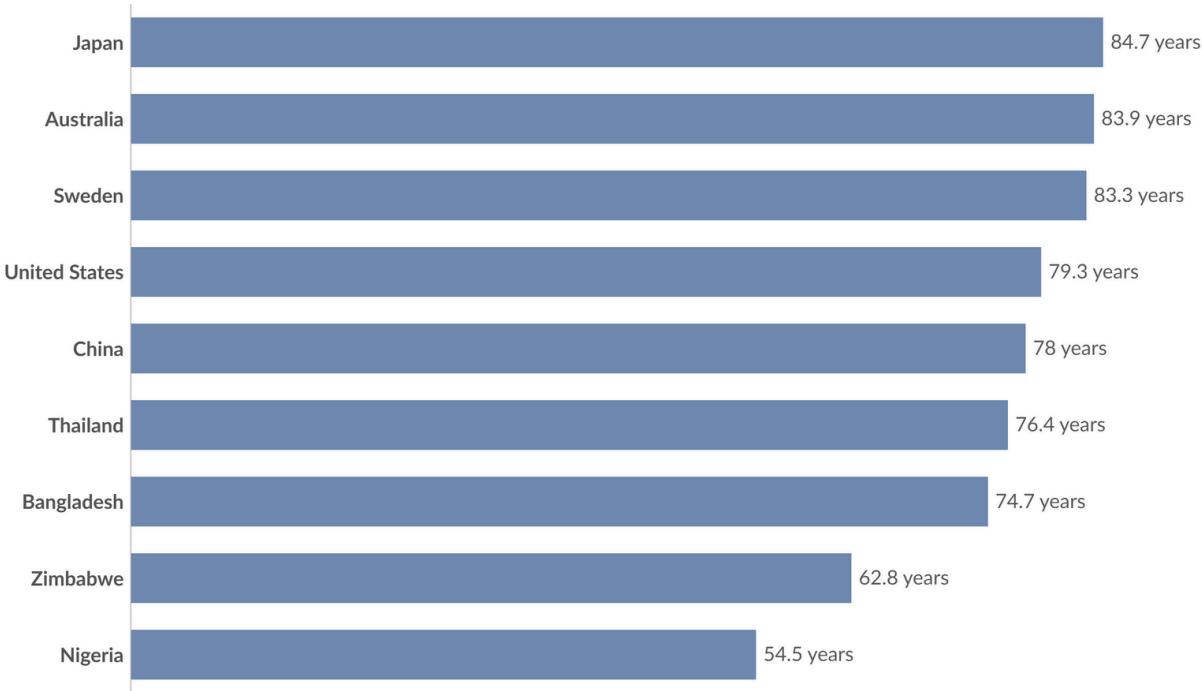
10 of 120 answers put the probability of D > probability of P

Income and life expectancy

Life expectancy at birth

Life expectancy, 2023

The period life expectancy¹ at birth, in a given year.



Data source: Human Mortality Database (2024); UN, World Population Prospects (2024) OurWorldinData.org/life-expectancy | CC BY

1. **Period life expectancy** Period life expectancy is a metric that summarizes death rates across all age groups in one particular year. For a given year, it represents the average lifespan for a hypothetical group of people, if they experienced the same age-specific death rates throughout their whole lives as the age-specific death rates seen in that particular year.
Learn more in our articles:
• [Life expectancy – what does this actually mean?](#)
• [Period versus cohort measures: what's the difference?](#)

Life expectancy is based on an estimate of the average age that members of a particular population group will be when they die

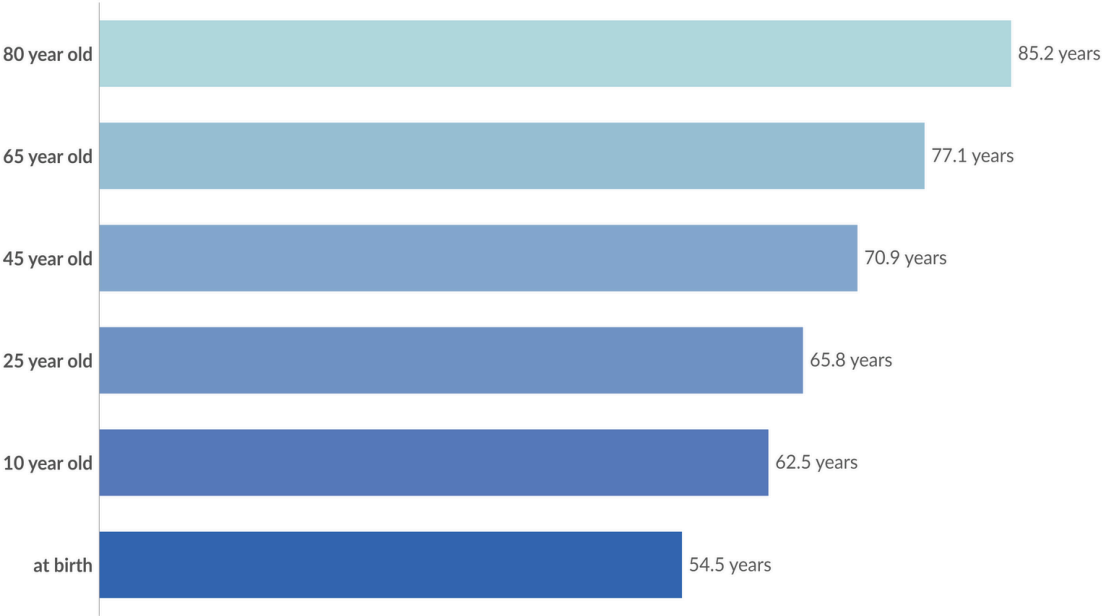
Source:
[Our World in Data](#)

Life expectancy by current age

Life expectancy for people of different ages, Nigeria, 2023



The total period life expectancy¹ for people who have reached a given age.



Data source: Riley (2005); Zijdeman et al. (2015); HMD (2024); UN WPP (2024); Human Mortality Database (2024); UN, World Population Prospects (2024)

OurWorldinData.org/life-expectancy | CC BY

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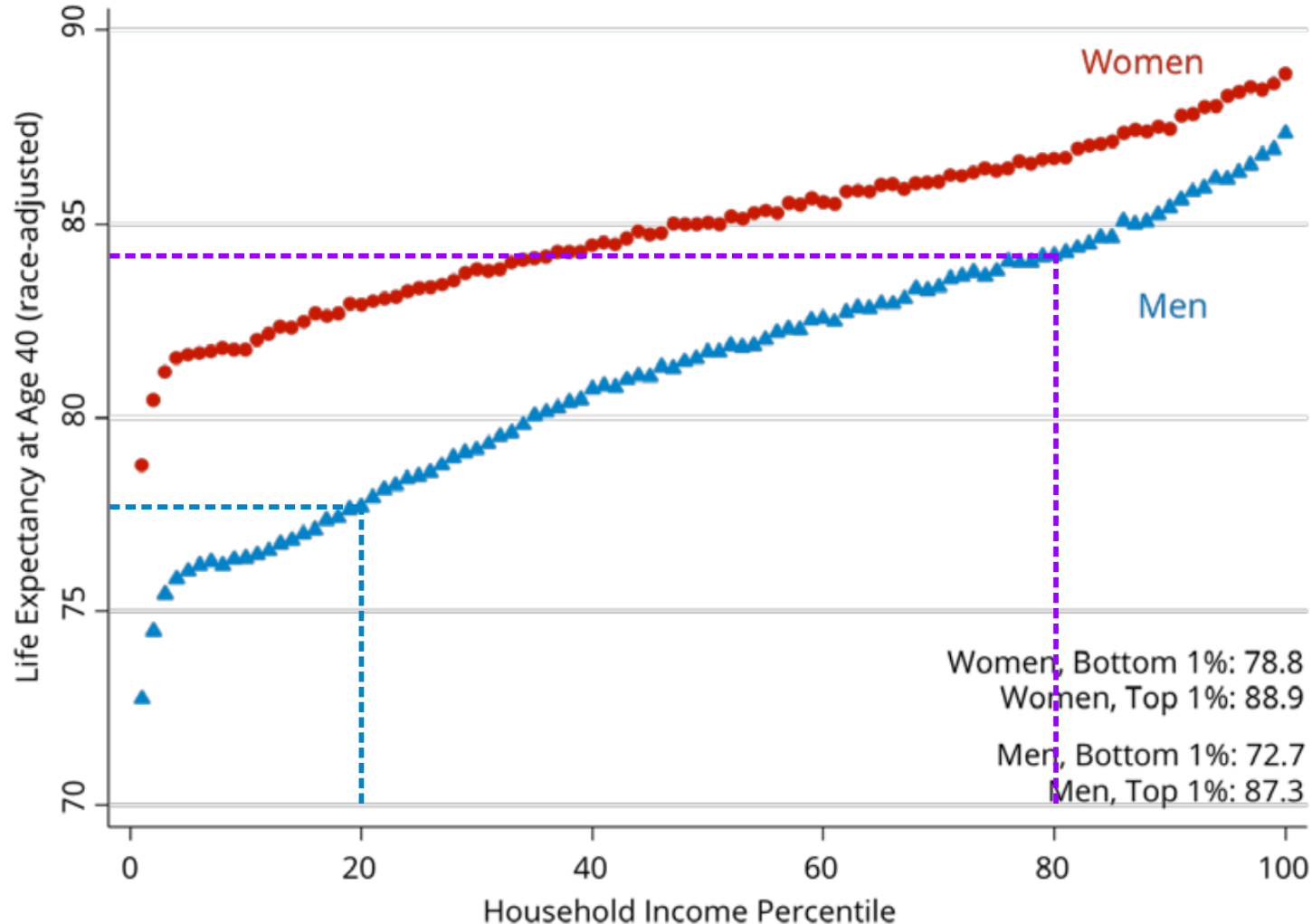
Guess the gap

US men, aged 40, 2001 - 2014

How many more years does a man in the richest 1% expect to live than a man in the poorest 1%?

The gap is 14.6 years

Figure 1: Life Expectancy vs. Income in the United States



Men, age 40

- Poorest 1%: 72.7 years
- Richest 1%: 87.3 years

Women: a gap of 10.1 years

Y : life expectancy at 40

X : income percentile

Each dot is $E[Y \mid X = x]$

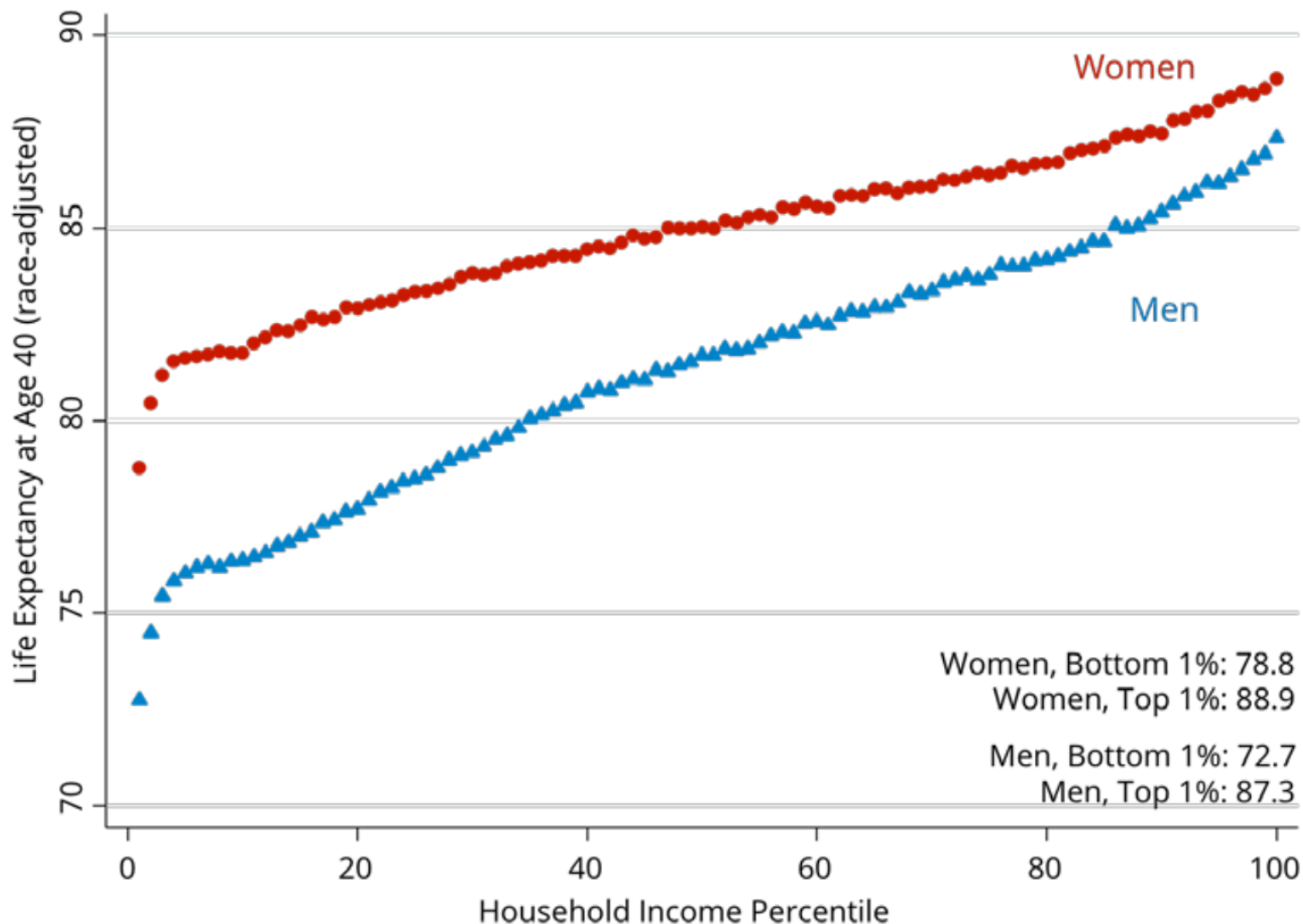
$$E[Y \mid X = \text{p20}] \approx 77$$

$$E[Y \mid X = \text{p80}] \approx 84$$

Source: Chetty et al. (2016)

We also condition on sex here

Figure 1: Life Expectancy vs. Income in the United States



Y : life expectancy at 40

X : income percentile

Z : sex

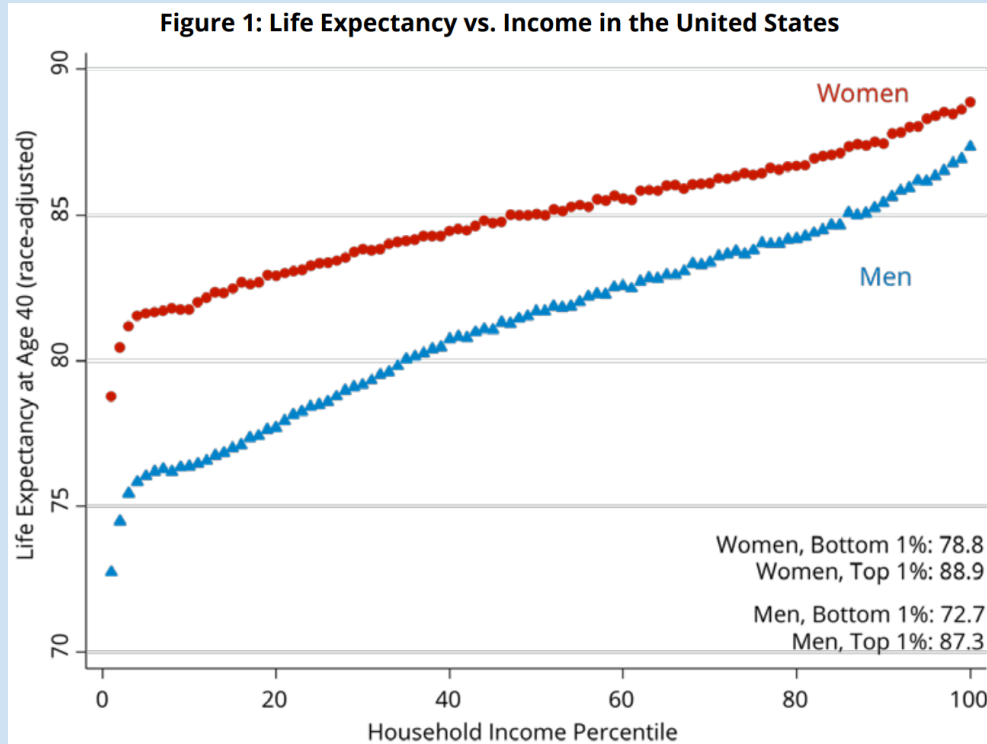
Each line is

$$E[Y \mid X = x, Z = z]$$

Source: [Chetty et al. \(2016\)](#)

Unconditional vs. conditional expectation

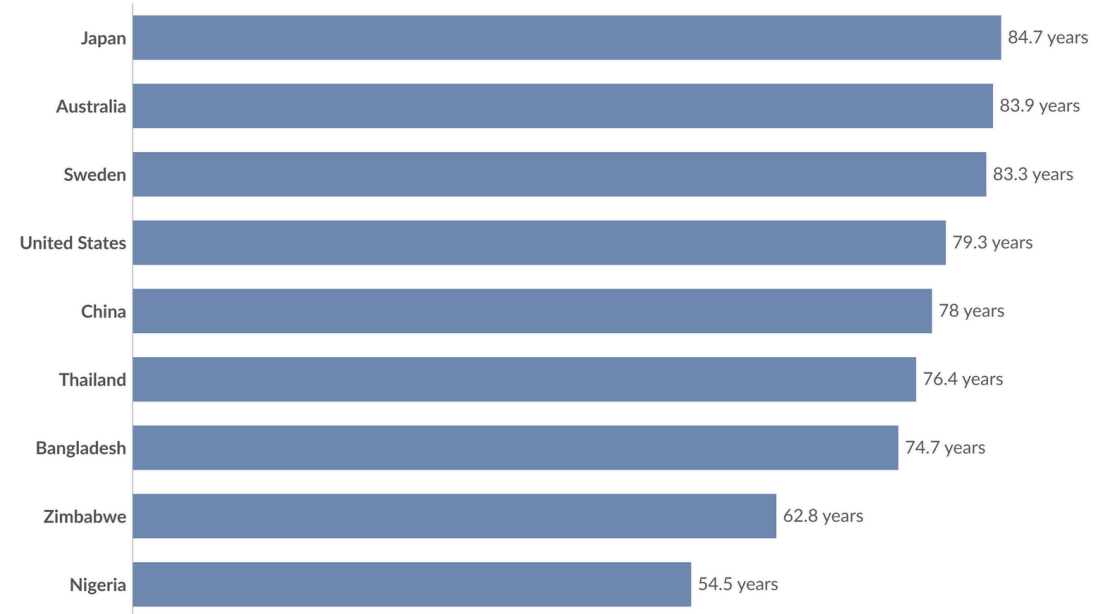
$E[Y]$ hides a lot of heterogeneity that we can see with $E[Y|X]$!



Life expectancy, 2023

The period life expectancy¹ at birth, in a given year.

Our World
in Data



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OurWorldinData.org/life-expectancy | CC BY

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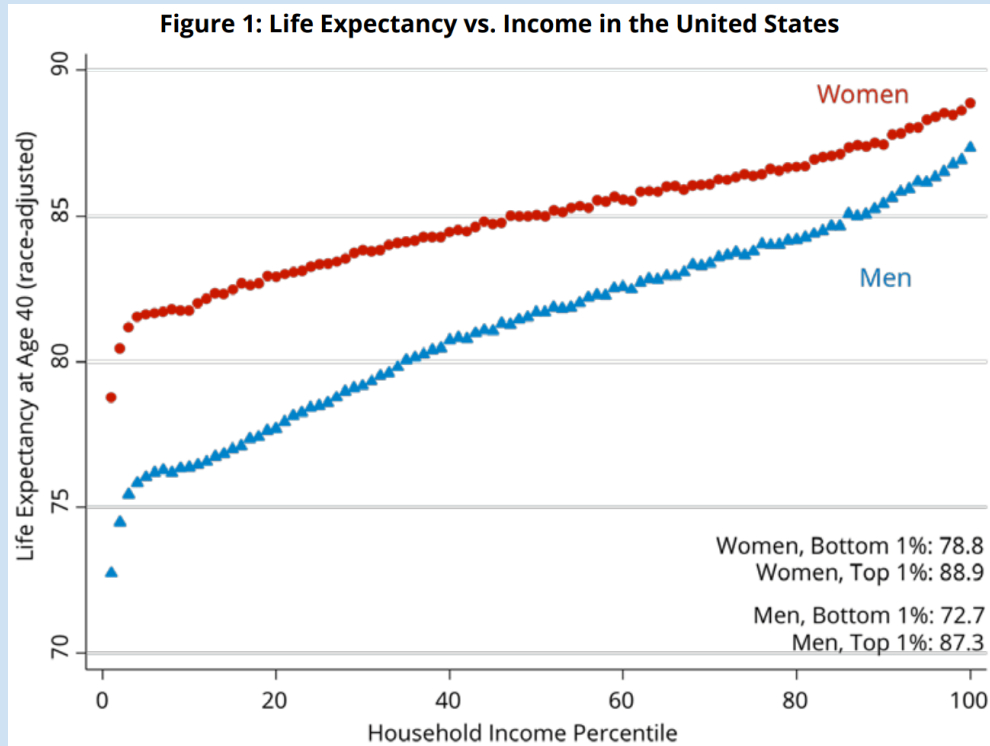
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Looking beyond averages

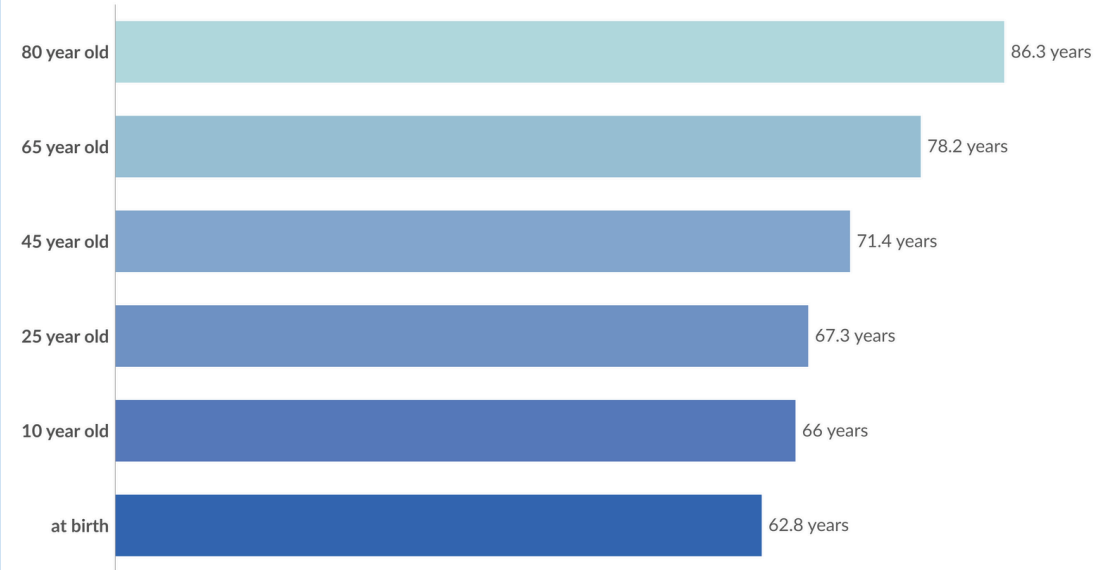
Low-income men in the US have $\approx$ LE as Zimbabwean men!



Life expectancy for people of different ages, Zimbabwe, 2023

The total period life expectancy¹ for people who have reached a given age.

Our World in Data



Data source: Riley (2005); Zijdemann et al. (2015); HMD (2024); UN WPP (2024); Human Mortality Database (2024); UN, World Population Prospects (2024)

OurWorldinData.org/life-expectancy | CC BY

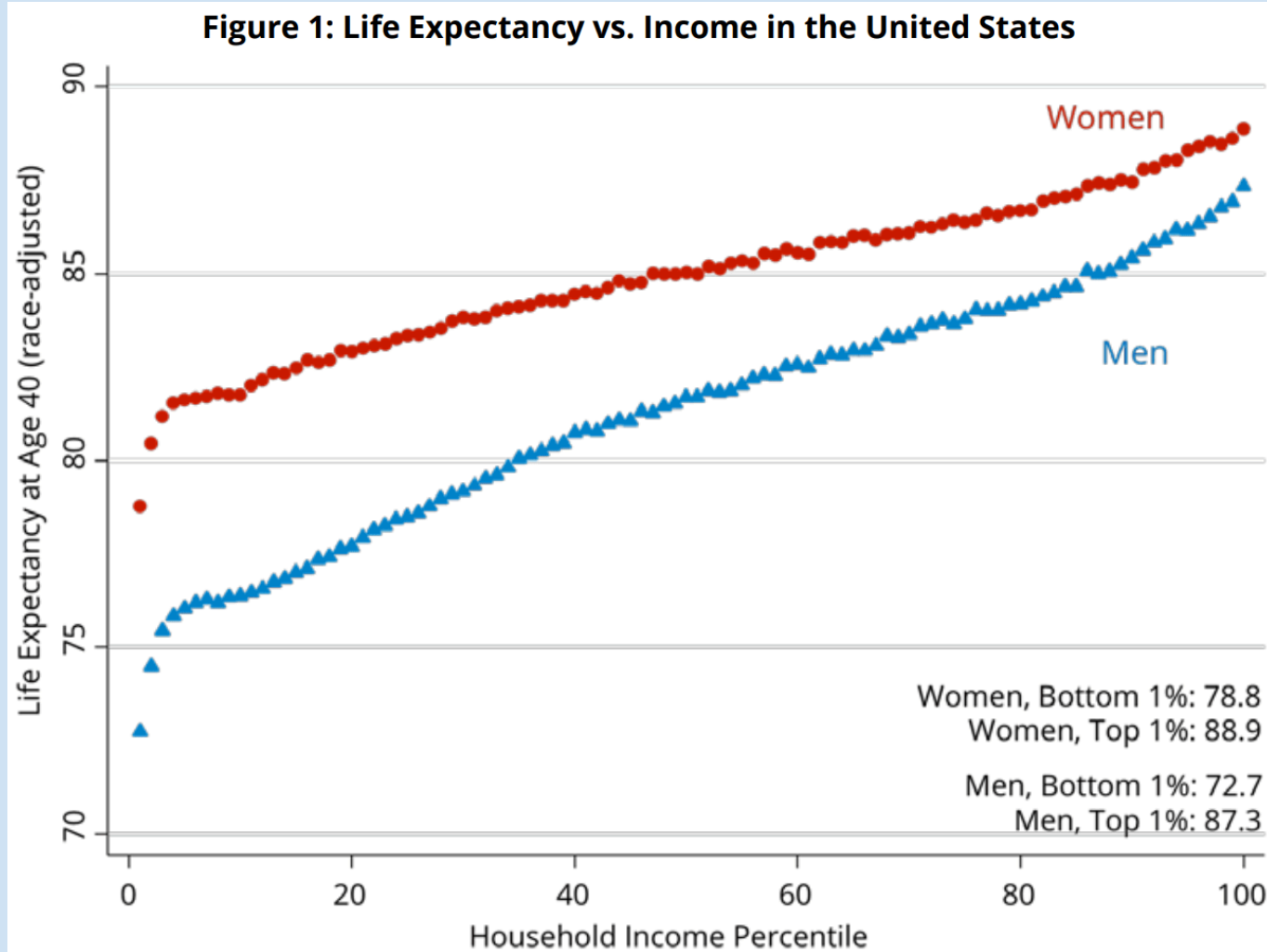
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Is this a causal relationship?

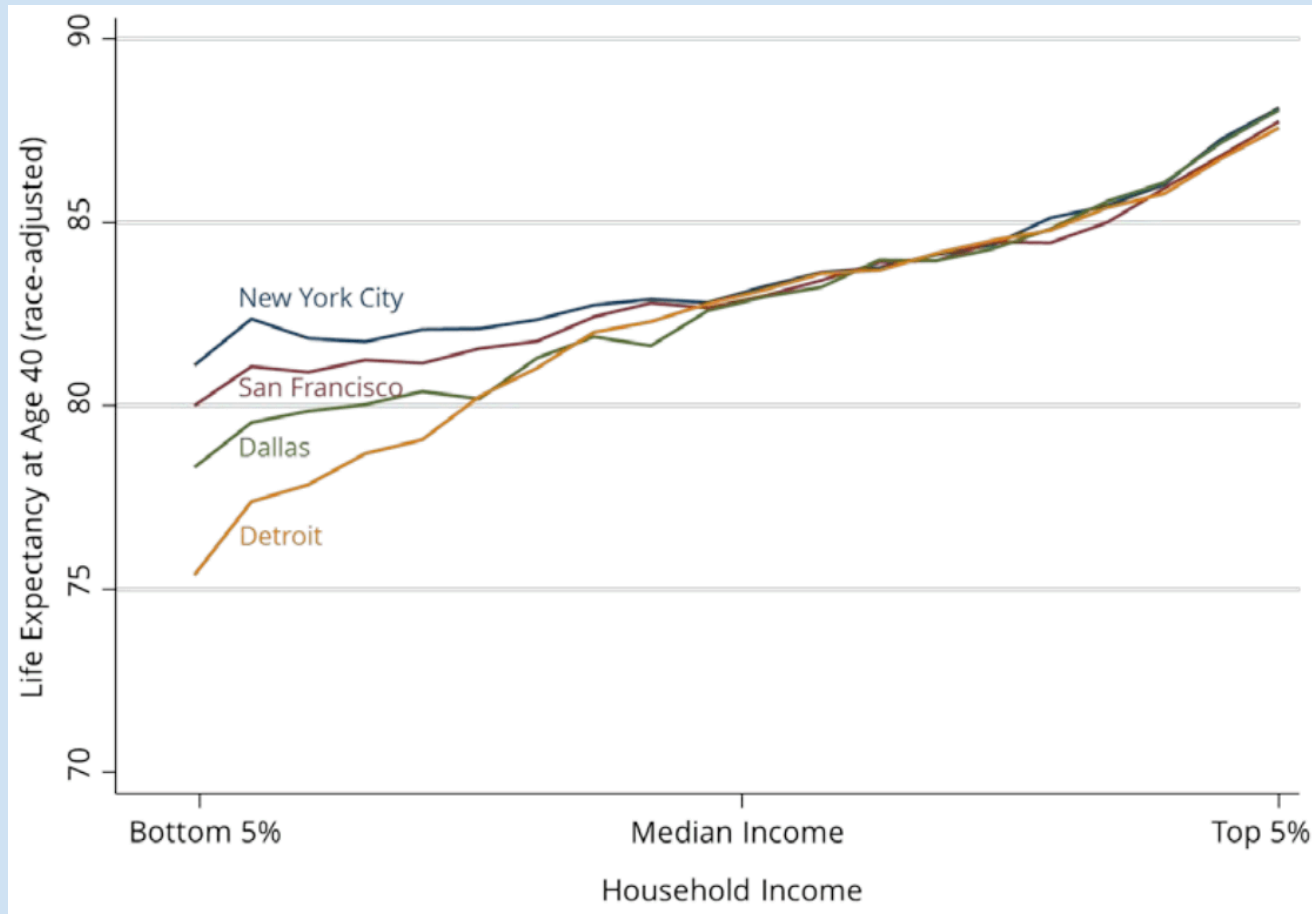


Does this suggest that giving more money to those at the bottom of the income distribution would $\uparrow$ their life expectancy?

Source: [Chetty et al. \(2016\)](#)

Descriptives can be incredibly powerful!

Place matters!



It's better to be poor in New York than in Detroit

Source: [Chetty et al. \(2016\)](#)

Some take-aways

- A conditional expectation is a function: one average for every value of X
- Averaging hides the ends: 14.6 years, inside one country
- Next week: the correlation gallery and where a covariance comes from